

Exploratory Data Analysis Report

Geldium Credit Delinquency Prediction Dataset

Prepared for:	Tata iQ Analytics Team & Geldium Decision-Makers
Dataset:	Delinquency_prediction_dataset.xlsx
Records Analysed:	500 customers × 19 variables
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1. Dataset Overview

The Geldium dataset contains **500 customer records** across **19 variables** covering demographic information, credit behaviour, financial indicators, and a 6-month payment history. The target variable is **Delinquent_Account** (binary: 1 = delinquent, 0 = non-delinquent). The dataset is moderately imbalanced, with 420 non-delinquent accounts (84%) and 80 delinquent accounts (16%).

Variable Categories

Category	Variables	Count
Demographic	Age, Location	2
Financial	Income, Loan_Balance, Debt_to_Income_Ratio	3
Credit Profile	Credit_Score, Credit_Utilization, Credit_Card_Type	3
Behavioural	Missed_Payments, Account_Tenure, Employment_Status	3
Payment History	Month_1 ... Month_6 (On-time / Late / Missed)	6
Identifier / Target	Customer_ID, Delinquent_Account	2

2. Data Quality Observations

The dataset exhibits several quality issues that must be addressed before predictive modelling. Missing values are present in two critical financial fields, and the Employment_Status column contains inconsistent categorical labels — a common consequence of manual data entry. A small number of records also show physically impossible Credit_Utilization values. These issues, if unaddressed, could introduce systematic bias into any downstream model.

2.1 Missing Values

Column	Missing Count	% Missing	Criticality
Income	39	7.8%	HIGH
Loan_Balance	29	5.8%	MEDIUM
Credit_Score	2	0.4%	LOW

2.2 Inconsistent Categorical Labels

The **Employment_Status** column contains six distinct values due to inconsistent capitalisation and abbreviations: *Employed*, *employed*, and *EMP* all represent the same category. This creates three separate buckets for what should be a single group (totalling 240 records, 48% of the dataset), which will mislead any model that treats them as distinct classes.

Raw Value	Count	Intended Category
Employed	82	Employed
employed	77	Employed ← inconsistent
EMP	81	Employed ← inconsistent
Unemployed	93	Unemployed
retired	87	Retired
Self-employed	80	Self-employed

2.3 Anomalous Values

Credit_Utilization should theoretically be capped at 1.0 (100%). However, **4 records** exceed this boundary (max observed: 1.026), suggesting data entry errors or accounts with balance overruns. These records require review before modelling.

3. Missing Data — Handling Strategy

Missing Data Issue	Chosen Method	Justification
Income (39 missing, 7.8%)	Median imputation (by Employment_Status group)	Income is right-skewed; group-level medians reduce bias introduced by extreme earners.
Loan_Balance (29 missing, 5.8%)	Predictive imputation (regression using Income, Debt_to_Income_Ratio)	Loan balance is strongly correlated with income and debt ratio; regression preserves these relationships.
Credit_Score (2 missing, 0.4%)	Mean imputation	Negligible volume; mean substitution has minimal impact on overall distribution.

Additional cleaning steps required: Standardise Employment_Status labels to four categories (Employed, Unemployed, Retired, Self-employed) using a lookup/replace operation. Cap Credit_Utilization values above 1.0 at 1.0 or flag for manual review.

4. Key Risk Factors & Detected Patterns

4.1 Top Predictors of Delinquency

#	Risk Indicator	Observation	Why It Matters
1	Missed_Payments	Range: 0–6; mean = 2.97. High frequency directly drives the target variable.	Historically the strongest behavioural signal for credit default risk.
2	Credit_Utilization	Mean = 49.1%; max = 102.6%. High utilisation signals borrower stress.	Utilisation >70% is a standard industry flag for elevated risk.
3	Debt_to_Income_Ratio	Range: 10%–55%; mean = 29.9%. Ratios >40% indicate repayment difficulty.	A high DTI constrains discretionary income and increases default likelihood.
4	6-Month Payment History (Month_1–6)	Roughly equal split across On-time/Late/Missed each month — no clear single peak.	Sequential late/missed payments reveal deteriorating repayment patterns.
5	Employment_Status	Unemployed = 93 records (largest single category after cleaning).	Unemployment directly reduces repayment capacity and elevates default risk.

4.2 Unexpected / Noteworthy Findings

- **Credit Score does not strongly separate delinquent from non-delinquent accounts.** Mean credit score for delinquent accounts (591) is actually slightly *higher* than for non-delinquent (575). This counter-intuitive finding warrants further investigation — it may reflect score inflation, recency lags in bureau reporting, or sample bias.
- **Income alone is not a reliable delinquency predictor.** Delinquent customers have a marginally higher mean income (\$113,902) than non-delinquent customers (\$107,307), suggesting that absolute income level is less important than how income is managed relative to debt obligations.
- **Payment history shows no worsening trend over 6 months.** The proportion of 'Missed' and 'Late' payments remains roughly consistent across Month_1 to Month_6, implying the dataset does not capture a deteriorating cohort — rather, a cross-section of customers with chronic payment difficulties.
- **Class imbalance (84% / 16%) must be addressed.** Without correction (e.g., SMOTE oversampling, class-weight adjustments), any predictive model will be biased toward predicting non-delinquency, producing misleadingly high accuracy but poor recall for the delinquent class.

5. Summary & Recommendations

The Geldium dataset provides a solid foundation for building a delinquency prediction model, but several data quality issues must be resolved before modelling begins. Missing values in Income and Loan_Balance (the two most financially significant predictors) should be imputed using group-level medians and regression-based methods respectively, while the Employment_Status column requires standardisation to eliminate duplicate categories. The most promising predictive variables are Missed_Payments, Credit_Utilization, and Debt_to_Income_Ratio, with 6-month payment history providing valuable sequential behavioural signal. Notably, Credit_Score and Income exhibit unexpected relationships with the target variable and should not be relied upon as primary predictors without further investigation. The 16% delinquency rate also means class imbalance must be explicitly managed during model training to ensure adequate sensitivity for identifying at-risk customers.

Priority	Action Required
1 — Immediate	Standardise Employment_Status labels (Employed / Unemployed / Retired / Self-employed)
2 — Before Modelling	Impute Income (group median) and Loan_Balance (regression); mean-impute 2 Credit_Score records
3 — Before Modelling	Cap Credit_Utilization at 1.0; flag 4 outlier records for review
4 — During Modelling	Apply SMOTE or class-weight balancing to address 84/16 class split
5 — Feature Engineering	Create aggregate payment-history feature (e.g., count of Missed across 6 months)